# Andrew Marion

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## EDUCATION

**KENNESAW STATE UNIVERSITY (KSU)** | Marietta, GA — Anticipated Graduation May 2027

• *Bachelor of Science in Mechanical Engineering, Aerospace Engineering Minor* — GPA: **3.81/4.0**

## RELEVANT EXPERIENCE

**NASA Glenn Research Center** | Cleveland, OH — August 2026 – December 2026

*UAV Flight Control Intern, Diagnostics and Controls Branch*

- Implement custom MATLAB/Simulink flight control algorithms on Pixhawk flight computers running PX4 autopilot for a 12 ft wingspan, 16-fan distributed electric propulsion (DEP) UAV based on NASA's SUSAN Electrofan configuration
- Develop fault-tolerant control allocation algorithms in MATLAB that redistribute thrust after fan or thrust-vectoring failures, evaluated across 8,000 Monte Carlo fan-failure cases
- Model UAV flight dynamics and powertrain behavior in MATLAB/Simulink and X-Plane for control law development

**Textron GSE** | Cartersville, GA — June 2026 – August 2026

*Mechanical Engineering Intern, New Product Development*

- Designed 3 assemblies and 15 parts in Siemens NX with production drawings for electric ground support equipment, including the MX4 baggage tractor hinged door and 660 belt loader access doors
- Ran motion simulation and prototype tests, including parking brake sensor placement and door kinematics, verifying function and reducing design rework before production release

## RESEARCH EXPERIENCE

**KSU Advanced Aerodynamics and Propulsion Lab** | Marietta, GA — September 2023 – Present

*Undergraduate Research Assistant*

**Numerical Investigation of Compound Delta Wings** — September 2023 – Present

- Executed 140+ RANS CFD simulations in ANSYS Fluent across Mach numbers and angles of attack, increasing lift-to-drag ratio 7.24–13.57% over the baseline wing in the transonic regime

**Multi-Fidelity Assessment of Control Surface Modeling with FlightStream** — March 2025 – Present

- Evaluated mid-fidelity control surface modeling in FlightStream with Oklahoma State's S2FAR Lab and Siemens
- Validated FlightStream against CFD, flight test, and WindShape wind tunnel data; published at AIAA SciTech 2026

**Aerodynamic Characterization of a Blended Wing Body Using FlightStream** — January 2026 – Present

- Benchmarked FlightStream against a 42-case transonic RANS sweep (Mach 0.6–0.8); FlightStream ran 5× faster with lift coefficient within 0.013 in the linear range

## PUBLICATIONS

- Marion, A., Sharma, G., et al., "Aerodynamic Characterization of a Blended Wing Body Using a Surface Vorticity Solver," accepted for presentation at AIAA SciTech 2027
- Comer, A., Atkinson, Z., Marion, A., et al., "Multi-Fidelity Assessment of Mid-Fidelity Control Surface Modeling with FlightStream," AIAA SciTech 2026. doi.org/10.2514/6.2026-2810
- Plus 4 more papers accepted to AIAA SciTech 2027 and a first-author 2025 AIAA Region II Student Conference paper (doi.org/10.2514/6.2025-97857)

## LEADERSHIP AND COMPETITION TEAMS

**Kennesaw Rocketry Team** | Marietta, GA — May 2025 – June 2026

*Vice President (December 2025 – June 2026); Liquid Propulsion Lead (May 2025 – December 2025)*

- Designed a 500 N nitrous oxide/ethanol liquid rocket engine; led propellant and injector trade studies in CAD and CFD

**Aerial Robotics Competition Team** | Marietta, GA — September 2023 – April 2025

*PADA Team Lead*

- Led 20+ members through design, manufacturing, and flight testing of a powered autonomous delivery aircraft (PADA), placing 10th of 19 teams in the Advanced Class at SAE Aero Design 2024
- Configured and tuned a Matek H743 flight controller running ArduPilot for autonomous flight of an inherently unstable flying wing that released from a carrier aircraft mid-flight and navigated to target landing coordinates

## TECHNICAL SKILLS

**CFD and Aerodynamics:** ANSYS Fluent, FlightStream, SolidWorks Flow Simulation, OpenVSP, ParaView, RANS, LES
**Controls and Programming:** MATLAB/Simulink, Python, R, LaTeX, PX4, ArduPilot, Pixhawk, X-Plane, Monte Carlo
**CAD:** Siemens NX, SolidWorks, AutoCAD, ANSYS SpaceClaim, Engineering Drawings
**Manufacturing:** 3D Printing, Laser Cutting, Composite Layup, Lathe/Mill, CNC Machining, Foam Cutting